

5(a)	<i>any two from</i> sequence of DNA nucleotides ; forms part of a DNA molecule / length of DNA / AW ; coding for a polypeptide ; A protein	2
5(b)	interphase ; I phases of interphase	1
5(c)(i)	two phosphates added to existing phosphate ; <i>base</i> = adenine ; <i>sugar</i> = ribose ;	3
5(c)(ii)	purine ;	1

Question	Answer	Marks
5(d)	<i>any three from</i> to produce, many / AW, new blood cells ; <i>idea that</i> large numbers are required to, transport oxygen / maintain immune system ; produce genetically identical cells ; loss of genetic material may not allow function / cells maintain function / cells are able to function ; replacement of, old / damaged / dead, cells ; AVP ; e.g. to maintain healthy numbers of cells red blood cells have short life	3